Satellite gravimetry · GRACE and GRACE-FO

Weighing Delhi's water from space

How twin satellites measure groundwater loss in north-west India using gravity

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Earth imagery: NASA Blue Marble and Black Marble. Spacecraft model: NASA/JPL-Caltech. Size and orbit not to scale.

10.3
km³ per year

of groundwater is being lost from Punjab, Haryana, Delhi and Rajasthan, Apr 2002 to Jul 2026.

Our estimate: −2.34 ± 0.05 cm of water per year over 4,38,913 km², from 259 months of satellite gravity data. About 3,25,676 litres every second.

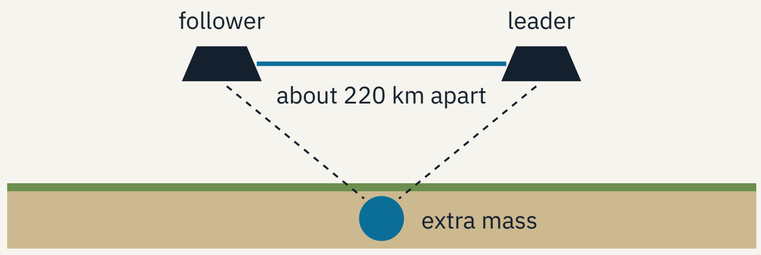
1 Gravity changes with water

Gravity is slightly stronger above more mass. Water is heavy, so a full aquifer pulls a little harder than an empty one. For a wide flat layer of water of thickness *h* (the Bouguer slab):

$$\Delta g = 2\pi G \rho h \quad \Rightarrow \quad \mathbf{0.419}\ \mu\text{Gal per cm of water}$$

Earth's gravity is about 980,000,000 µGal, so the signal is tiny but measurable.

2 Two satellites feel the pull



The pair orbits about 500 km up and tracks its separation to about a micron. Extra mass ahead pulls the leader first, so the gap grows, then shrinks as the follower catches up. A month of such measurements gives one gravity map of the Earth.

Schematic, not to scale. Distances from NASA/JPL.

3 From gravity to groundwater

GRACE weighs *all* the water in a region (total water storage, TWS). We subtract the parts a land model can estimate:

$$\Delta GWS = \Delta TWS - (\text{soil moisture} + \text{snow} + \text{canopy water})$$

TWS: JPL mascons RL06.3Mv04, checked against CSR mascons. Land model: NASA GLDAS-2.1 Noah. All values are anomalies relative to 2004–2009, area-weighted on the sphere. Missing months are left empty, never filled in.

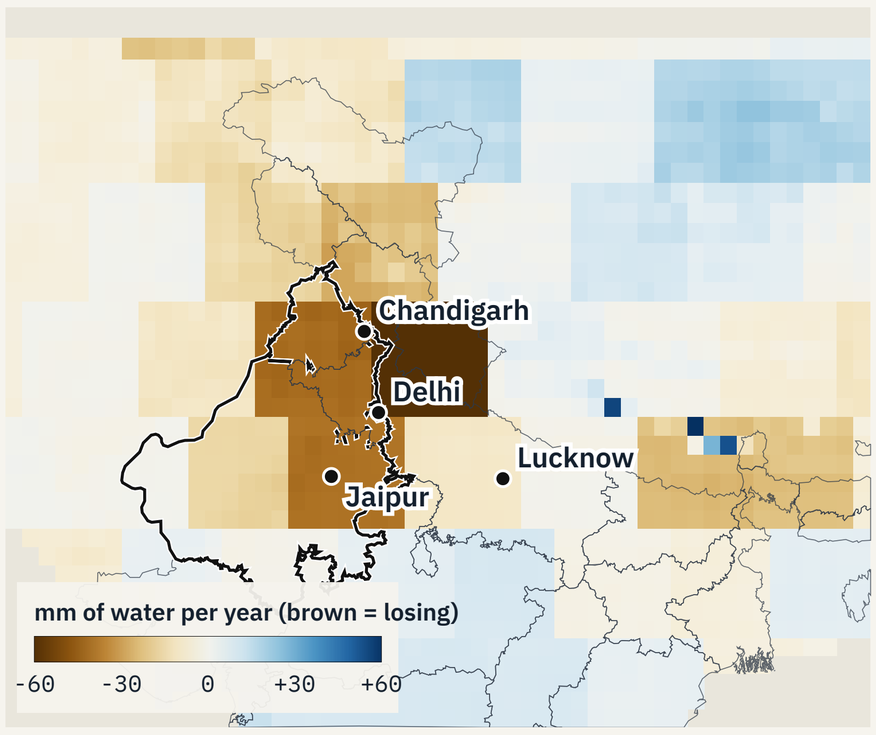


Figure 1. Groundwater trend, Apr 2002 to Jul 2026, in mm of water per year (JPL mascons minus GLDAS). Thick outline: our analysis region. The square blocks are real: GRACE resolves about 300 km, so each 3° mascon moves as one. State boundaries: DataMeet (CC BY 4.0).

4 Results

NW India trend, Apr 2002 to Jul 2026	cm/yr	km³/yr
Groundwater, JPL (headline)	**−2.34 ± 0.05**	**−10.3**
Groundwater, JPL with gain factors	−2.28	−10.0
Groundwater, CSR solution	−2.48	−10.9
Total water storage, JPL	−2.15 ± 0.06	−9.4

km³/yr = Σ (trend in m/yr × cell area in m² × fraction of cell in region) ÷ 10⁹. 33 of 294 months have no satellite solution.

Compared with the first study

Rodell et al. (2009) reported −4.0 ± 1.0 cm/yr (−17.7 ± 4.5 km³/yr) for the same states, August 2002 to October 2008. Our fit over those months gives −2.59 ± 0.17 cm/yr (−11.4 km³/yr): the same direction, but smaller. We use newer mascon data and did not tune anything to match.

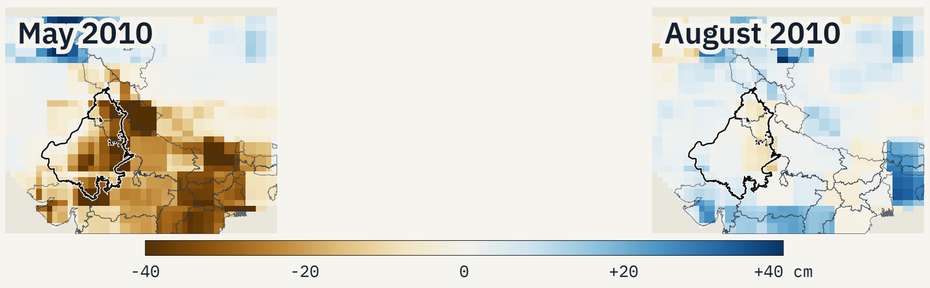


Figure 3. The monsoon in the data: total water storage before (May 2010) and during (August 2010) the monsoon, cm of water relative to 2004–2009.

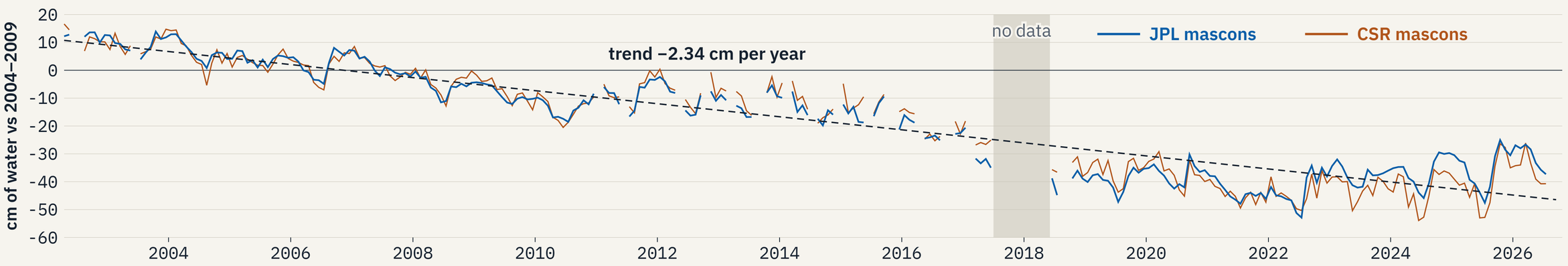


Figure 2. Groundwater storage anomaly for north-west India from two independent GRACE solutions. Every year the monsoon refills the ground; underneath, the level falls. Dashed line: least-squares trend fitted together with annual and semi-annual cycles. Grey band: no data between the end of GRACE (2017) and the start of GRACE-FO (2018).

5 What to keep in mind

- GRACE sees features about 300 km across; our region is only about a dozen independent measurements.
- Groundwater is what is left after subtracting a model, so it carries the model's errors. Rivers and reservoirs are not removed.
- The ± values are formal fit errors and a lower bound; the spread between solutions is the more honest guide.
- A straight line simplifies the record: the fall is steady to about 2017 and flatter, with large swings, afterwards.
- The satellites show *that* water is lost, not who used it. Rodell et al. (2009) attribute the loss mainly to pumping for irrigation.

Data and references

- JPL GRACE/GRACE-FO Mascons, CRI filtered, RL06.3Mv04. NASA PO.DAAC. doi:10.5067/TEMSC-3JC634
- CSR GRACE/GRACE-FO RL06.3 Mascons. Univ. of Texas. doi:10.15781/cgq9-nh24
- GLDAS Noah L4 monthly 0.25° V2.1. NASA GES DISC. doi:10.5067/SXAVCZFAQLNO
- Rodell, Velicogna & Famiglietti (2009). Nature 460, 999–1002. doi:10.1038/nature08238
- Watkins et al. (2015). JGR Solid Earth. doi:10.1002/2014JB011547
- Save, Bettadpur & Tapley (2016). JGR Solid Earth. doi:10.1002/2016JB013007

All numbers and figures are computed from these data products. Not endorsed by NASA or JPL.

Explore it yourself

Scan for the interactive site: play the map month by month, spin India in 3D, and see every equation and data source.

`sankalpabiswas40.github.io/grace-groundwater`